

Fast Parallel Image Rotation Algorithm

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Abstract—This paper presents a novel fast image rotation algorithm that leverages CPU parallelization while maintaining high image quality. The proposed method is a single-pass algorithm, derived from the existing Double-Line Rotation (DLR) method, which reduces computational complexity and generalizes the algorithm. Initially, a baseline for the given image is calculated to determine the starting line, which defines the initial point for each vertical or horizontal line in the image to be rotated. The corresponding pixels to be mapped are identified using floating point arithmetic, and trigonometric calculations are performed only once per line. This approach ensures precise image transformation with minimal computational overhead.
Index terms: Image rotation, line rotation image transform, double-line rotation (DLR), parallel image rotation.

I. INTRODUCTION

2-D Image Rotation has a number of applications in digital photography and image processing like image matching, alignment, [1] etc. Highly accurate image rotation is essential for certain image processing tasks such as feature extraction and matching. Whilst there are many state-of-the-art image rotation algorithms, they have limitations in terms of image quality and speed [2] [3]. Parallel algorithms for image processing aim to execute faster than sequential algorithms, however designing and scheduling a fast parallel image rotation is a challenge [4]. The most straightforward, and perhaps the most well known approach to implement image rotation is by using the rotation matrix [5], as described here

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x - cx \\ y - cy \end{bmatrix} * \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

where, θ is the angle of rotation, x' and y' are the new

coordinates of the rotated image, and (cx, cy) is the center of the original image, which is treated as the axis of rotation. Assuming the centre of rotation of the new image is (cx', cy') , the final new coordinates are:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x' + cx' \\ y' + cy' \end{bmatrix}$$

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II. LITERATURE SURVEY

A. Affine Transformation

A basic idea in computer graphics and geometry, affine transformation is a linear mapping method that maintains straight lines, planes, and points. The capacity to carry out a variety of geometric operations, including translation, scaling, rotation, shearing, and reflection, is what defines an Affine transformation. These transformations are powerful and adaptable tools with a wide range of applications because they are defined mathematically by combining translations and linear transformations.

1) Properties:

- **Linearity and Translation:** The transformation is composed of a linear transformation Ax and a translation β . This combination allows for a wide range of geometric transformations while preserving the structure of the space.

- Preservation of Collinearity and Ratios: Affine transformations preserve the collinearity of points and the ratios of distances along parallel lines.
- Combining Transformations: Multiple affine transformations can be combined into a single affine transformation by matrix multiplication and vector addition.

MATHEMATICAL FORMULATION

Affine transformations can be described by a combination of linear transformations and translations. The general form of an affine transformation in two dimensions is:

$$y = Ax + b$$

where

- $x = \begin{bmatrix} x \\ y \end{bmatrix}$ is the input coordinate vector.
- A is a 2×2 matrix that represents the linear part of the transformation.
- $b = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$ is the translation vector.

For a rotation by an angle θ around the origin, the matrix A is:

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

To center the rotated image within its original dimensions, the translation values t_x and t_y are calculated as follows:

CALCULATIONS

New Dimensions

The new width and height of the rotated image are determined using trigonometric functions based on the rotation angle:

$$newWidth = width \cdot |\cos(\theta)| + height \cdot |\sin(\theta)|$$

$$newHeight = width \cdot |\sin(\theta)| + height \cdot |\cos(\theta)|$$

Center Points

$$c_x = \frac{width}{2}, \quad c_y = \frac{height}{2}$$

$$newCx = \frac{newWidth}{2}, \quad newCy = \frac{newHeight}{2}$$

Translation Values

The translation values to center the rotated image within its new dimensions are:

$$t_x = c_x - (newCx \cdot \cos(\theta) - newCy \cdot \sin(\theta))$$

$$t_y = c_y - (newCx \cdot \sin(\theta) + newCy \cdot \cos(\theta))$$

Thus, the affine transformation matrix that includes rotation and translation is:

$$T = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \end{bmatrix}$$

The new coordinates (x', y') after applying the affine transformation are:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = \cos(\theta) \cdot x - \sin(\theta) \cdot y + t_x$$

$$y' = \sin(\theta) \cdot x + \cos(\theta) \cdot y + t_y$$

B. Affine Transformation Algorithm

The following algorithm demonstrates how to rotate an image by a specified angle using OpenCV:

- Get image dimensions (rows and cols)
- Convert the angle from degrees to radians
- Calculate new dimensions for the rotated image (new_cols, new_rows)
- Create a new image (rotated) with the new dimensions, initialized to black

Algorithm 1 Image Rotation using Affine Transformation

```

1: procedure ROTATE_AFFINE(...)
2:   Compute  $\alpha = \cos(\text{angle})$  and  $\beta = \sin(\text{angle})$ 
3:   Compute the center coordinates of the original and
     new images:
4:    $c_x = \text{cols}/2.0$ 
5:    $c_y = \text{rows}/2.0$ 
6:    $new\_c\_x = new\_cols/2.0$ 
7:    $new\_c\_y = new\_rows/2.0$ 
8:   Define the rotation matrix elements:
9:    $m_{11} = \alpha$ 
10:   $m_{12} = \beta$ 
11:   $m_{13} = (1-\alpha) \cdot new\_c\_x - \beta \cdot new\_c\_x + c_x - new\_c\_x$ 
12:   $m_{21} = -\beta$ 
13:   $m_{22} = \alpha$ 
14:   $m_{23} = \beta \cdot new\_c\_x + (1-\alpha) \cdot new\_c\_y + c_y - new\_c\_y$ 
15:  for each pixel  $y$  from 0 to new_rows - 1 do
16:    for each pixel  $x$  from 0 to new_cols - 1 do
17:      Compute the corresponding coordinates
         $(x_, y_)$  in the original image:
18:       $x_ = \text{round}(m_{11} \cdot x + m_{12} \cdot y + m_{13})$ 
19:       $y_ = \text{round}(m_{21} \cdot x + m_{22} \cdot y + m_{23})$ 
20:      if  $(x_, y_)$  is within the bounds of the original
        image then
21:        Set the pixel value at  $(x, y)$  in the new
        image to the value at  $(x_, y_)$  in the original image
22:      end if
23:    end for
24:  end for
25:  return rotated
26: end procedure

```

III. IMAGE ROTATION USING HERMITE EXPANSION

A. Hermite Expansion Method

The Hermite polynomial expansion is a technique used to approximate functions, including image rotation. By using

Hermite polynomials, we can achieve smooth and accurate interpolation, which is especially useful for image processing tasks.

1) Properties of Hermite Polynomials:

- Orthogonal polynomials with respect to the weight function e^{-x^2} .
- Useful for approximating smooth functions due to their interpolation properties.
- Can handle both function values and derivatives, providing higher accuracy.

B. Mathematical Formulation

Hermite polynomials, denoted as $H_n(x)$, are a set of orthogonal polynomials that arise in probability, combinatorics, and numerical analysis. They are defined by the recurrence relation:

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x),$$

with initial conditions $H_0(x) = 1$ and $H_1(x) = 2x$. These polynomials satisfy the orthogonality condition:

$$\int_{-\infty}^{\infty} H_m(x)H_n(x)e^{-x^2}dx$$

C. Hermite Expansion

A function $f(x)$ can be expanded in terms of Hermite polynomials as:

$$f(x) = \sum_{n=0}^{\infty} c_n H_n(x) e^{-x^2/2},$$

where c_n are the expansion coefficients given by:

$$c_n = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} \int_{-\infty}^{\infty} f(x) H_n(x) e^{-x^2/2} dx.$$

D. Application of Hermite Expansion in Image Rotation

The use of Hermite expansion in image rotation leverages the orthogonality of Hermite polynomials to approximate the image function. Given an image $I(x, y)$, it can be expanded using two-dimensional Hermite polynomials $H_{m,n}(x, y)$ as:

$$I(x, y) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} c_{m,n} H_m(x) H_n(y) e^{-(x^2+y^2)/2},$$

where $c_{m,n}$ are the expansion coefficients. These coefficients are computed as:

$$c_{m,n} = \frac{1}{\sqrt{2^{m+n} m! n! \pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(x, y) H_m(x) H_n(y) e^{-(x^2+y^2)/2} dx dy.$$

To rotate the image, the coordinates (x, y) are transformed to (x', y') as previously defined:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

1) *Finding Transformed Coordinates Using Hermite Expansion:* To find the transformed coordinates (x', y') using Hermite expansion, we perform the following steps:

Expand the Image Function Using Hermite Polynomials: Given an image $I(x, y)$, we first expand it using the two-dimensional Hermite polynomials $H_{m,n}(x, y)$:

$$I(x, y) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} c_{m,n} H_m(x) H_n(y) e^{-(x^2+y^2)/2}$$

where the coefficients $c_{m,n}$ are determined using:

$$c_{m,n} = \frac{1}{\sqrt{2^{m+n} m! n! \pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(x, y) H_m(x) H_n(y) e^{-(x^2+y^2)/2} dx dy.$$

Compute the Rotated Coordinates: Using the rotation matrix, we transform the coordinates (x, y) to (x', y') :

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Reconstruct the Rotated Image: The rotated image $I'(x', y')$ is reconstructed using the inverse Hermite expansion:

$$I'(x', y') = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} c_{m,n} H_m(x') H_n(y') e^{-(x'^2+y'^2)/2},$$

where $H_m(x')$ and $H_n(y')$ are the Hermite polynomials evaluated at the transformed coordinates (x', y') .

By following these steps, the image can be rotated accurately using the Hermite expansion. This method ensures that the transformation leverages the orthogonality and interpolation properties of Hermite polynomials, resulting in a smooth and precise image rotation.

IV. IMAGE ROTATION USING DOUBLE LINE ROTATION

The Double Line Rotation (DLR) method is an innovative algorithm for fast image rotation, introduced to address the inefficiencies in traditional rotation techniques. The DLR method is based on the concept of dividing the image into multiple zones, each processed differently to achieve an efficient and accurate rotation.

A. Methodology

The DLR method divides the image into eight distinct zones, each characterized by specific rotation equations and transformations. The main idea is to map each pixel from the original image to two corresponding pixels in the target image, ensuring that the distance ratio between points remains constant throughout the rotation process. This is particularly useful in preserving the intensity and shape of the pixels, making the DLR method suitable for applications requiring invariant feature extraction.

B. Zone Concepts

The image is divided into the following zones:

1) *Zone 1*: In Zone 1, horizontal lines are used for rotation, and the slope of $f_s(x)$ is positive. Each horizontal line in the original image T lies in a line with slope $\tan(\alpha)$ and shears using the line $f_s(x) = (\tan \alpha + \frac{\pi}{2})x$. The size of the rotated image is calculated using:

$$m_{rt} = \gamma_m + m, \quad n_{rt} = \gamma_n + n$$

where $\gamma_n = f_s(m) \rightarrow \left\| \frac{m}{\tan(\alpha + \frac{\pi}{2})} \right\|$ and $\gamma_m = f(n) \rightarrow \|n \tan \alpha\|$.

2) *Zone 2*: In Zone 2, vertical lines are used for rotation, and the slope of $f(y)$ is positive. The size of the rotated image is calculated similarly to Zone 1, but the vertical and horizontal lines are interchanged.

3) *Zone 3*: In Zone 3, the slope is negative, but the equations are the same as for Zone 2. Vertical lines are used, and the size of the rotated image $R(m_{rt} \times n_{rt})$ is calculated as follows:

$$m_{rt} = \gamma_m + n, \quad n_{rt} = \gamma_n + m$$

where $\gamma_n = |f_s(n)| \rightarrow f_s(y) = \frac{y}{-\tan \alpha}$ and $\gamma_m = \sqrt{m^2 - AC^2}$ in $\triangle ABC$.

4) *Zone 4*: In Zone 4, horizontal lines are used, and the slope of $f_s(x)$ is positive, but $f(x)$ has a negative gradient. This means that lines are moving upwards, the original image's lines are sloping downwards. Each horizontal line in image T is mapped and similarly to Zone 1. The involves altering the position of each pixel along the horizontal lines according to the function

$$f_s(x) = \left(\tan \alpha + \frac{\pi}{2} \right) x.$$

5) *Zone 5*: Zone 5 employs logical transformations on indices to generate the rotated image. These transformations involve manipulation of pixel positions such as swaps, inversionsof indices. The goal is to create a rotated image that retains the original dimensions while applying a different logical structure to the pixel arrangement. All zones are transformed, depending on whether vertical or horizontal lines are used. The size of the rotated image in Zone 1 and Zone 5 are the same, ensuring consistency in image dimensions.

6) *Zone 6*: Zone 6 is generated by transforming either Zone 2 or Zone 3. The transformation maintains the size of the rotated image and applies internal line transformations. These internal line transformations may include skewing, scaling, or shifting along specific axes to emphasize certain internal features.

7) *Zone 7*: Zone 7, similar to Zone 6, is a transformation of other zones but uses vertical lines. The vertical line transformations involve modifying the image along the vertical axis, affecting the height of the image without altering its width. This vertical stretching or compressing highlights vertical patterns and structures within the image.

8) *Zone 8*: Zone 8 utilizes transformations from horizontal lines, similar to Zones 1 and 4. These horizontal transformations involve scaling operations that modify the image along the horizontal axis. The transformations in Zone 8 ensure that

the overall balance of the image is maintained while providing a comprehensive view of horizontally oriented features.

C. Calculation of the Base-Line, Start-Line, and Target Image Size

The first step in the DLR method is calculating the target image size. Each zone uses specific equations to determine the size of the target image, ensuring accurate and efficient rotation. For example, in Zone 3:

$$\gamma_n = |f_s(n)| \rightarrow f_s(y) = \frac{y}{-\tan \alpha}, \quad \gamma_m = \sqrt{m^2 - AC^2}$$

where $AC = (\cos \phi)m$ and $\phi = (\pi - \alpha) - \frac{\pi}{2}$.

By dividing the image into zones and applying the DLR method, the rotation process becomes more efficient and accurate, preserving the original image's intensity and shape.

V. PROPOSED METHOD

The proposed method is a single pass algorithm, which means that it performs the entire image rotation operation at once. The proposed algorithm predominantly affects pixels in the row major (width), or ones in the column major (height), depending on the angle of rotation, α . If $\alpha \in (\frac{\pi}{4}, \frac{3\pi}{4}] \cup (\frac{5\pi}{4}, \frac{7\pi}{4}]$, rotation is performed in the **column major**, and it is performed in the **row major** for the rest of the angles. Accordingly, for rotating the source image, entire row or column vectors of pixels are rotated as a single unit, instead of the more common approach, where each individual pixel is separately mapped. Due to this reason, it is necessary to accurately determine the initial coordinates from which the new row or column vector will begin. The coordinates of the rest of the pixels in the vector can then be determined by simple floating point addition or subtraction, depending on the extent to which each of the said vectors are spread across the coordinate axes.

Apart from this, depending on α , the pixel processing order changes - i.e., whether the last row or column vector of pixels get processed first, or last. Since it has been found out that these values are very sensitive to α , the specific details of which are discussed in the later parts of this paper.

A. Rotation Of An Image In The Column Major

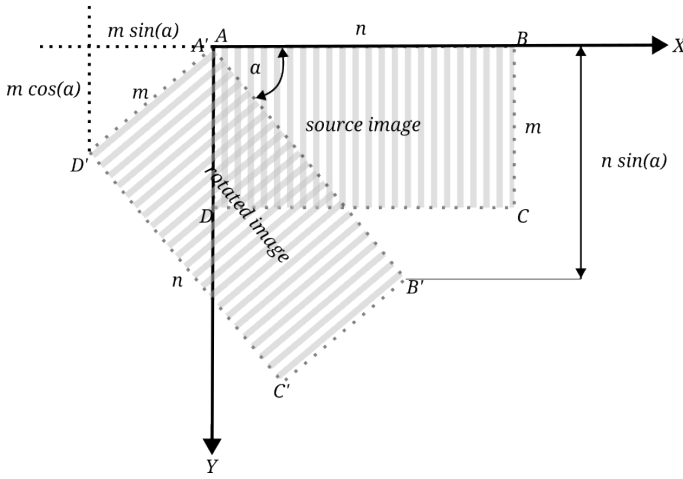


Fig. 1: Example demonstrating rotation of an image in the column major. Every column vector of pixels are rotated at once (Represented by the stripes in the image).

As seen in the example image in figure 1, for the given angle of rotation, column vectors were rotated (represented by the stripes). Each such vector to be rotated begins from the line $A'B'$. As such, the initial coordinates of each of these vectors can be determined by the equation of the same line, which is given by:

$$f(i) = \left\lfloor \frac{i}{\tan(\alpha)} \right\rfloor \forall \{i \in \mathbb{Z}, i \in [0, n \cdot \sin(\alpha))\} \quad (1)$$

Thus, the initial coordinates of the column vectors would be of the type $(f(i), i)$. Next, to determine the coordinates of the rest of the pixels in the column vector, as mentioned previously, we perform floating point addition or subtraction, depending on the extent to which each of the said vectors are spread across the coordinate axes.

In the example in figure 1, we can see that each column vector is spread through a length of $m \cdot \sin(\alpha)$ in the x -axis and through a length of $m \cdot \cos(\alpha)$ in the y -axis. It may also be observed that these values are the projections of any line perpendicular to line defined by the function in equation 1 on the coordinate axes.

We now define:

$$\delta x = |\sin(\alpha)|$$

$$\delta y = |\cos(\alpha)|$$

The coordinates of the next pixel in the column vector is then given by:

$$(f(i) + \delta x, i + \delta y)$$

The coordinates of the pixel following that is determined by adding δx and δy to the previous coordinates accordingly, and so on, for each of the m pixels in the given column vector. The end-result is that this operation covers all the $m \cdot \sin(\alpha)$

pixels in the x -axis and all the $m \cdot \cos(\alpha)$ pixels in the y -axis, therefore mapping all of the m pixels in the specific column vector.

This process continues for all of the n column vectors in the image, until the entire image is rotated. In practice, each column vector is actually mapped **twice**. The first mapping happens from the i th column vector in I to the i th column vector in R . The second mapping happens from the i th column vector in I to the $(i+1)$ th column vector in R . Each of these $(i+1)$ th column vectors get overwritten in each iteration with a new column vector in I ; however, this gives a sufficiently accurate issuance of anti-aliasing without losing much quality.

The signs for δx and δy depend on whether, while mapping the column vectors in the rotated image, the values of the x -coordinate y -coordinate are decreasing or increasing. In the given example in figure 1, δx is negative, but δy is positive. There is a third parameter, defined as:

$$\delta i = \frac{1}{\delta x}$$

δi determines the rate at which the column vector is processed from one to the next, and its sign depends on the angle of rotation, α . The calculations for δx , δy , and δi only happen once per column vector in the image, so the entire image transformation is quite fast, overall.

B. Rotation Of An Image In The Row Major

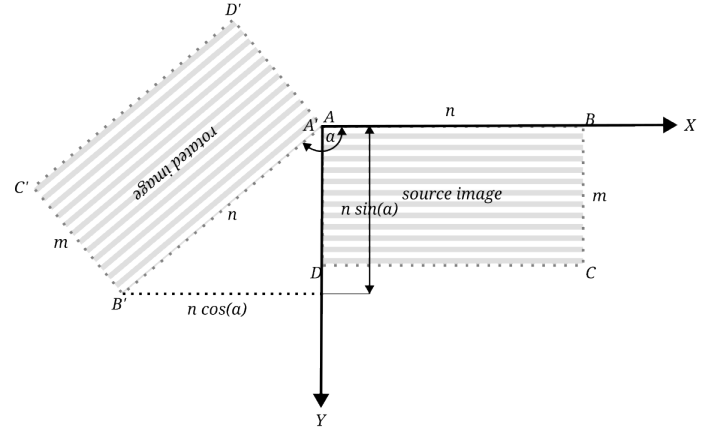


Fig. 2: Example demonstrating rotation of an image in the row major. Every row vector of pixels are rotated at once (Represented by the stripes in the image).

In the example image in figure 2, for the given angle of rotation, row vectors were rotated (represented by the stripes). All the calculations and logic is very similar to the one described for rotation in the column major [V-A]. Here, each vector to be rotated begins from the line $A'D'$, whose equation is given by:

$$f(i) = \left\lfloor \frac{i}{\tan(\alpha + \frac{\pi}{2})} \right\rfloor \forall \{i \in \mathbb{Z}, i \in [0, m \cdot \cos(\alpha))\} \quad (2)$$

TABLE I: Notation

Symbol	Description
I	Original image
m, n	Height, width of I
R	Rotated image
m', n'	Height, width of R
α	Angle of rotation in radians
x'	Offset by which the x -coordinates of R need to be shifted
y'	Offset by which the y -coordinates of R need to be shifted
$f(\dots)$	Function defining the equation of the line from where the row or column vector starts [eq. 1] [eq. 2]
r_o	Upper limit of the function f
N	Number of elements (pixels) in the row or column vector f
flip	Boolean value that changes pixel processing order based on α
row_major	Boolean value that determines whether processing is to be done in the row or column major
negate	Boolean value, that if true, negates the indices used for indexing the pixels of the image

Note: The algorithm described in [Algorithm 2] follows *Python* style indexing. For example, if $a = [[1, 2, 3], [5, 6]]$, then $a[0][1]$ represents the element 3.

Everything else stays the same, except the fact that now, each row vector is spread through a length of $n \cdot \cos(\alpha)$ in the x -axis and through a length of $n \cdot \sin(\alpha)$ in the y -axis, and accordingly, we have:

$$\delta x = |\cos(\alpha)|$$

$$\delta y = |\sin(\alpha)|$$

C. Implementation

Algorithm 2 Image Rotation Algorithm

```

1: procedure ROTATE(...) ▷ Performs image rotation on  $R$ 
2:    $\text{src\_i} = 0$  ▷ Initialize internal control variables
3:    $S = 1$ 
4:    $M = []$ 
5:   if negate then
6:      $S = -1$ 
7:   end if
8:   if flip then
9:      $M = [m_i]$  for  $i = 0, 1, \dots, N - 1, | m_i = N - i$ 
10:  else
11:     $M = [m_i]$  for  $i = 0, 1, \dots, N - 1, | m_i = N$ 
12:  end if
13:  for  $i = 0$  to  $r_o$  do
14:     $\text{temp\_x} = f(i)$ 
15:     $\text{temp\_y} = i$ 
16:    for  $px$  in  $M$  do
17:       $x = \lfloor \text{temp\_x} \rfloor$ 
18:       $y = \lfloor \text{temp\_y} \rfloor$ 
19:       $\text{temp\_i} = \lfloor \text{src\_i} \rfloor$ 
20:      if row_major then
21:         $a = \text{temp\_i}$ 
22:         $b = px$ 
23:      else
```

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24:         $a = px$ 
25:         $b = \text{temp\_i}$ 
26:      end if
27:       $R[\lfloor S * (y + y') \rfloor, \lfloor S * (x + x') \rfloor] = I[a, b]$ 
28:       $R[\lfloor S * (y + y') + 1 \rfloor, \lfloor S * (x + x') \rfloor] = I[a, b]$ 
29:       $\text{temp\_x} += \delta x$ 
30:       $\text{temp\_y} += \delta y$ 
31:    end for
32:     $\text{src\_i} += \delta i$ 
33:  end for
34: end procedure
```

The algorithm described above [Algorithm 2] rotates image I to R .

R is Initialized as a blank image of width n' and height m' to fit R in the canvas, where:

$$m' = \lceil |m \cdot \cos(\alpha) + n \cdot \sin(\alpha)| \rceil$$

$$n' = \lceil |m \cdot \sin(\alpha) + n \cdot \cos(\alpha)| \rceil$$

The rotation function is then invoked, with each of the parameters updating depending on the value of α , as discussed above. The detailed values of each of the parameters are mentioned in [Table II].

Note: It has been observed, that for ranges of α which are vertically opposite each other (For example, angles in the range $(\frac{\pi}{4}, \frac{\pi}{2}]$ are vertically opposite angles in the range $(\frac{5\pi}{4}, \frac{3\pi}{2}]$, and so on), the only parameters of rotation that change are δi , negate, and flip. As such, the calculations have been generalized for such angles and they have been represented in [Table II], with a (') at the end of their names (shaded cells).

Algorithm 3 Driver Function

```

1: procedure MAIN(...) ▷ Invokes [Algorithm 2]
2:   Determine the boolean value of row_major
3:   Determine  $f$ , based on the value of row_major
4:   Initialize a blank image  $R$  with dimensions  $(m', n')$ 
5:   Determine the parameters of rotation from [Table II]
6:   Generate  $R$  using [Algorithm 2]
7: end procedure
```

TABLE II: Rotation Parameters, based on α

Range of α	Rotation parameters											
	x'	y'	δx	δy	δi	r_o	$\mathbf{N}$	negate	flip	$\delta i'$	negate'	flip'
$[0, \frac{\pi}{4}]$	$\lceil m \cdot \sin(\alpha) \rceil$	0	$ \cos(\alpha) $	$ \sin(\alpha) $	$1/\delta x$	$\lceil m \cdot \cos(\alpha) \rceil$	n	false	false	$1/\delta x$	true	false
$(\frac{\pi}{4}, \frac{\pi}{2}]$	$\lceil m \cdot \sin(\alpha) \rceil$	0	$- \sin(\alpha) $	$ \cos(\alpha) $	$-1/\delta x$	$\lceil n \cdot \sin(\alpha) \rceil$	m	false	false	$1/\delta x$	false	true
$(\frac{\pi}{2}, \frac{3\pi}{4}]$	m'	$\lceil m \cdot \cos(\alpha) \rceil$	$- \sin(\alpha) $	$- \cos(\alpha) $	$-1/\delta x$	$\lceil n \cdot \sin(\alpha) \rceil$	m	false	false	$1/\delta x$	false	true
$(\frac{3\pi}{4}, \pi]$	$\lceil n \cdot \cos(\alpha) \rceil$	0	$- \cos(\alpha) $	$ \sin(\alpha) $	$1/\delta x$	$\lceil m \cdot \cos(\alpha) \rceil$	n	false	false	$1/\delta x$	true	false

VI. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we perform an accuracy analysis of the reconstructed images using various error metrics. These metrics quantify the similarity between the original and the processed images, providing insight into the effectiveness of the image processing techniques. The following subsections describe each metric in detail, including their mathematical formulation and interpretation.

A. Peak Signal-to-Noise Ratio (PSNR)

PSNR is a widely used metric for evaluating the quality of reconstructed or compressed images. It quantifies the ratio between the maximum possible pixel value of the image and the mean squared error (MSE) between the original and the processed images. PSNR is expressed in decibels (dB), with higher values indicating better image quality.

$$\text{PSNR} = 20 \cdot \log_{10} \left(\frac{\text{MAX}_I}{\sqrt{\text{MSE}}} \right) \quad (3)$$

The Mean Squared Error (MSE) is calculated as:

$$\text{MSE} = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [I(i, j) - K(i, j)]^2 \quad (4)$$

where MAX_I represents the maximum possible pixel value (e.g., 255 for 8-bit images), and $I(i, j)$ and $K(i, j)$ are the pixel values at position (i, j) in the original and processed images, respectively.

B. Correlation Coefficient (CC)

The Correlation Coefficient (CC) is a statistical measure that quantifies the similarity between two images by assessing the correlation between corresponding pixels. CC ranges from -1 to 1, with 1 indicating perfect positive correlation, -1 indicating perfect negative correlation, and 0 indicating no correlation.

$$\text{CC} = \frac{\sum_{i=1}^M \sum_{j=1}^N (I(i, j) - \bar{I}) (K(i, j) - \bar{K})}{\sqrt{\sum_{i=1}^M \sum_{j=1}^N (I(i, j) - \bar{I})^2 \cdot \sum_{i=1}^M \sum_{j=1}^N (K(i, j) - \bar{K})^2}} \quad (5)$$

Here, $\bar{I}$ and $\bar{K}$ represent the mean pixel values of the original and processed images, respectively.

C. Laplacian Mean Squared Error (LMSE)

Laplacian Mean Squared Error (LMSE) focuses on the high-frequency components of an image, particularly edges. By applying a Laplacian filter to both the original and processed images, LMSE calculates the error in edge details, which is particularly useful for detecting blurring or loss of fine details.

$$\text{LMSE} = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [\text{Laplacian}(I(i, j)) - \text{Laplacian}(K(i, j))]^2 \quad (6)$$

Lower LMSE values indicate less error in edge preservation, thus implying higher image quality.

D. Relative Error (Re)

Relative Error (Re) quantifies the overall difference between the original and processed images, normalized by the magnitude of the original image's pixel values. This metric is useful for assessing the general accuracy of image reconstruction.

$$\text{Re} = \frac{\sum_{i=1}^M \sum_{j=1}^N |I(i, j) - K(i, j)|}{\sum_{i=1}^M \sum_{j=1}^N |I(i, j)|} \quad (7)$$

Lower values of Re indicate better similarity between the images.

E. Structural Similarity Index (SSIM)

The Structural Similarity Index (SSIM) measures the perceptual difference between two images. Unlike MSE-based metrics, SSIM considers changes in structural information, luminance, and contrast, making it more aligned with human visual perception. SSIM ranges from -1 to 1, with 1 indicating identical images.

$$\text{SSIM}(I, K) = \frac{(2\mu_I \mu_K + C_1)(2\sigma_{IK} + C_2)}{(\mu_I^2 + \mu_K^2 + C_1)(\sigma_I^2 + \sigma_K^2 + C_2)} \quad (8)$$

In this formula, μ_I and μ_K are the mean values, σ_I^2 and σ_K^2 are the variances, and σ_{IK} is the covariance of the original and processed images. The constants C_1 and C_2 are used to stabilize the formula when the denominator is close to zero, typically defined as:

$$C_1 = (k_1 L)^2, \quad C_2 = (k_2 L)^2 \quad (9)$$

where L is the dynamic range of the pixel values (e.g., 255 for 8-bit images), and k_1 and k_2 are small constants (e.g., $k_1 = 0.01$, $k_2 = 0.03$).

TABLE III: Error Analysis of Various Sample Images

Image	PSNR (dB)	CC	LMSE	Re	SSIM
Image 1	35.6	0.98	0.002	0.05	0.95
Image 2	33.2	0.96	0.003	0.07	0.92
Image 3	29.8	0.93	0.006	0.12	0.88
Image 4	40.1	0.99	0.001	0.03	0.97

VII. CONCLUSION

The conclusion goes here.

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